

Ultimate Nutrition

D H E A

For many years DHEA, or dehydroepiandrosterone, was considered a "junk" hormone, since researchers could not attribute any specific function to it. In the past few years, DHEA's image problem has been totally refurbished. It is the most abundant hormone in humans and other mammals, and it plays a critical role in many different bodily functions. In fact, decrease in DHEA levels is being increasingly used as an indicator of reduced immune response, loss of libido and lack of a sense of general wellbeing. Clinical studies over the past ten years show that DHEA may be the most critical single chemical in predicting health and disease.

The body produces DHEA in the adrenal gland using cholesterol as the starting material. It is metabolized to androstenedione, which is converted directly to testosterone. Like testosterone (estrogen in women) and other critical hormones, DHEA levels peak in the second or third decade of life, after which the decline begins. In men, roughly one-half of DHEA is lost by age 40. Reduced levels of DHEA mean reduced amounts of androstenedione, testosterone and other steroid hormone metabolites. It is interesting to note that as cholesterol in general increases with age, DHEA levels decrease. As a result, DHEA deficiency is associated with an array of chronic conditions, such as obesity, diabetes, high blood pressure, coronary artery disease, and various immune deficiencies, including autoimmune diseases.

DHEA directly affects glucose metabolism. High blood glucose inhibits growth hormone (GH) and low blood glucose stimulates growth hormone. Stimulation of growth hormone is beneficial to DHEA, because it suppresses insulin. By the same token, fasting increases both GH and DHEA. Thus, the importance of DHEA in intense exercise and endurance training can hardly be overemphasized. In fact, Dr. William Regelson of Medical College of Virginia in his book *The Superhormone Promise* calls DHEA "the superstar of hormones" and "one of the most powerful tools" available for enhancing and extending life.

Clinical results show that oral supplementation with DHEA can have numerous beneficial effects including:

- Feelings of energy and wellbeing.
- Improved insulin sensitivity and glucose tolerance.
- Reduced risk of coronary artery diseases.
- Lower obesity/waist-to-hip ratio.
- Enhanced libido.
- Reduced depression and increases cognitive ability.

FREQUENTLY ASKED QUESTIONS

1. What is the Difference between DHEA and DHEA Sulfate?

The total DHEA in the blood is a combination of free DHEA and DHEA sulfate (DHEA-S), which is a DHEA molecule with a sulfur atom attached to it. Although no distinction is normally

made between DHEA and DHEA-S, it should be noted that DHEA-S functions as a ready reservoir of weakly bound DHEA.

2. Does Oral DHEA Supplementation Work?

Numerous studies, some of which are listed in the references above, have shown that oral DHEA works quite efficiently. Intake of as small an amount as 100 mg DHEA each day can result in remarkable increase in both physical and psychological wellbeing. It helps improve muscle mass and muscle strength is increased. Since it is a powerful hormone, in cases of any clinical conditions, it should be taken under clinical supervision and in consultation with your primary care physician.